

5	(a)	(Young modulus / $E =$) stress / strain	B1	[1]
	(b)	(i)		
		stress = F/A		
		or = $F/(\pi d^2/4)$		
		or = $F/(\pi d^2)$	M1	
		ratio = 4 (or 4:1)	A1	[2]
	(ii)	E is the same for both wires (as same material) [e.g. $E_P = E_Q$]	M1	
		strain = stress / E		
		ratio = 4 (or 4:1) [<i>must be same as (i)</i>]	A1	[2]